



IDBI INNOVATE 2026

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Team Details

- a. **Team name:** ASM FinTech Developer
- b. **Team leader name:** Abhinav Mukwane
- c. **Problem Statement:** Financial Health Score

Brief about the idea

India has 64 million MSMEs contributing ~30% of GDP, yet 80%+ lack formal credit access. New-to-Credit (NTC) and New-to-Bank (NTB) enterprises are routinely rejected despite genuine creditworthiness — simply due to missing paperwork.

FinRiskLensAI : AI-Powered MSME Financial Health Score Platform

- Aggregates alternate data streams: GST returns, Account Aggregator & UPI statements, ITR filings, Udyam registry (EPFO slot ready)
- Computes a multi-dimensional Financial Health Score (0-1000) across 6 weighted pillars using ML.NET (LightGBM model + rules engine)
- Generates a visual Financial Health Card with radar chart, trend graph, risk flags, and credit product recommendations
- Integrates with ULI (DSP API), OCEN 2.0 (LSP interface), and AA FIU consent framework for near real-time assessment
- Near real-time scoring: end-to-end score in under 10 seconds from consent trigger
- PFI-based explainability: every score carries a dimension-level reason - audit-ready for credit officers
- Recommends CGTMSE/MUDRA-backed products for mid-range scorers to maximise viable approvals
- Enables IDBI Bank to on-board credit-invisible MSMEs while strengthening portfolio quality and sector diversification

Opportunities

How different is it from any of the other existing ideas?

- Unlike CIBIL/Experian bureau scores, FinRiskLensAI works for NTC/NTB borrowers with zero credit history
- Unlike single-source tools, we unify 4 DPI data streams (GST + AA + UPI + ITR + EPFO) in one consent-based flow
- First solution combining ULI DSP-compatible API + OCEN 2.0 LSP + AA FIU consent flow in a single platform
- GST vs. bank-credit cross-validation as a proprietary fraud signal — no existing tool does this

How will it be able to solve the problem?

- Eliminates the 'no document, no credit' rejection cycle by using MSMEs' existing digital footprints as credit proxies
- Near real-time scoring replaces 2-3 week manual evaluation with a sub-10-second automated assessment
- PFI explainability gives credit officers dimension-level reasons — no black-box decisions
- Neutral baseline for NTC status prevents unfair penalization of first-time borrowers

USP of the proposed solution

- 6-dimensional score gives IDBI a far richer risk picture than any single metric (Revenue, Cash Flow, Transactions, Compliance, Stability, Debt)
- Built entirely on India's DPI stack — zero proprietary data dependency, fully consent-based and DPDP Act 2023 compliant
- Credit product recommender maps score band to CGTMSE/MUDRA schemes — maximizing both approval rate and safety
- ULI DSP-compatible API means any OCEN-compliant lender can plug in — IDBI's infrastructure becomes reusable fintech rails

List of features offered by the solution

Data Aggregation

- Consent-based multi-source pull: Udyam registry, GST API (GSTR-1/2A/2B/3B, ITC), ITR filings, AA bank & UPI statements (EPFO slot ready)
- GST vs. bank-credit cross-validation for fraud/misrepresentation detection

AI / ML Scoring Engine

- 6-Dimension Weighted Score (0-1000): Revenue Vitality (25%), Cash Flow Health (20%), Tx Trustworthiness (15%), Compliance (15%), Business Stability (15%), Debt Serviceability (10%)
- ML.NET LightGBM for tabular scoring + SSA time-series for 12-month trend + RandomizedPCA for cross-source anomaly detection
- PFI (Permutation Feature Importance): per-dimension reasons for every score - audit-ready

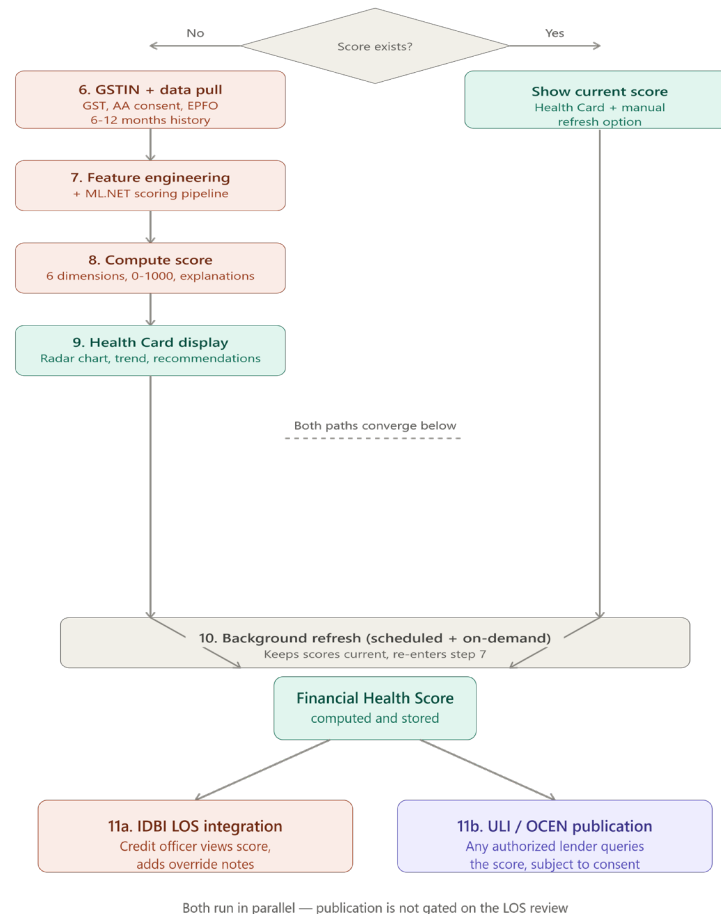
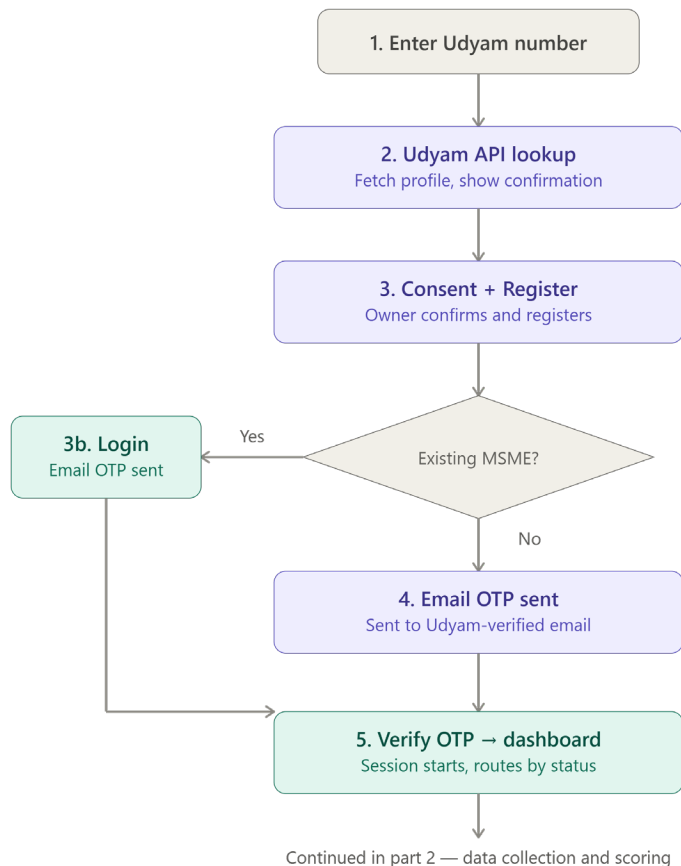
Financial Health Card UI

- Radar/spider chart for 6 dimensions, 12-month trend graph, green/yellow/red risk flags, top 3 strengths and risks
- Recommended credit products (CGTMSE, MUDRA, standard working capital) mapped to score band
- Downloadable PDF Health Card for branch-level use by IDBI credit officers

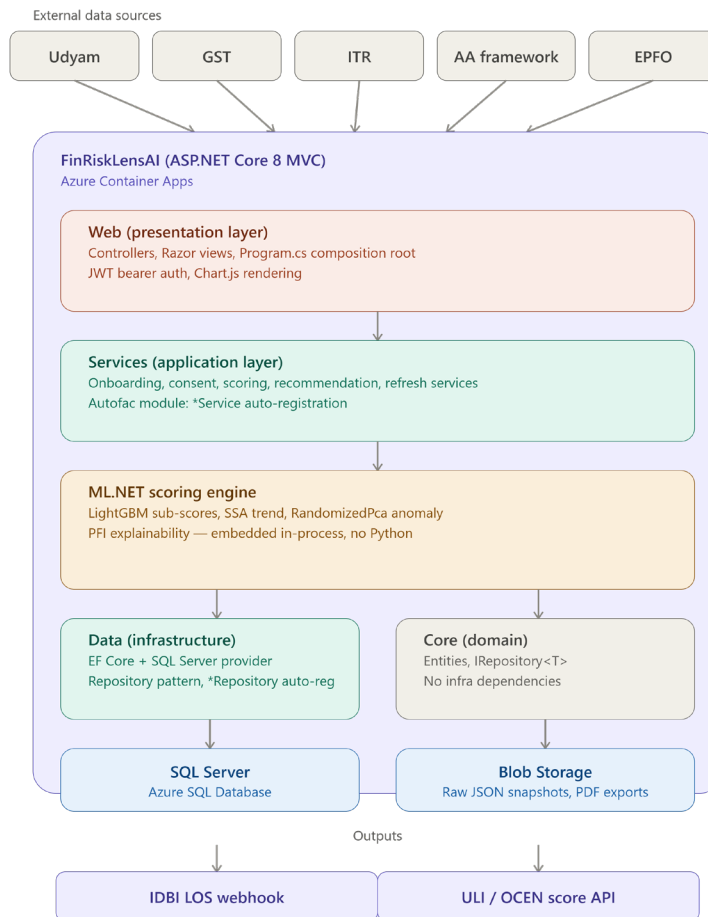
Integration & API

- ULI DSP-compatible REST API: any OCEN-compliant lender queries the score with mTLS + OAuth2
- IDBI LOS Integration: score + Health Card delivered inline to Loan Origination System
- AA FIU consent lifecycle: consent request, artefact verification, FIP data pull, auto-expiry
- On-demand refresh: a new GST filing or AA consent re-runs the blob-based scoring pipeline

Process flow diagram or Use-case diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution

Backend — Orchestration & API

- ASP.NET Core 8 MVC (C#): orchestration + Udyam/GST/ITR/AA (Finvu) connectors, ULI DSP, IDBI LOS webhook
- ML.NET (in-process): scoring, feature engineering & PFI explainability - no separate ML service
- Autofac: DI & module auto-registration | Serilog: structured file/console logging

AI / ML Stack

- ML.NET LightGBM: calibration model; final score = 0.6 rules engine + 0.4 ML model
- ML.NET SSA (Singular Spectrum Analysis): 12-month cashflow / revenue trend
- Permutation Feature Importance (PFI): per-dimension, audit-ready explanations
- ML.NET RandomizedPCA: 3-way GST vs ITR vs bank-credit anomaly / fraud signal

Data & Storage

- SQL Server + EF Core 8: MSME profiles, consent & OTP tables, audit trail (code-first migrations)
- Azure Blob Storage: per-MSME data folder (raw source JSON + result.json)

Infrastructure & DevOps

- ASP.NET Core 8 web app - Web + ML.NET in one deployable (modern monolith Architecture)
- Azure API Management (planned): ULI/OCEN gateway with mTLS + OAuth2
- GitHub: source control & CI | IIS / Azure App Service / AWS : hosting

Technologies to be used in the solution

Frontend

- ASP.NET Core MVC + Razor Views: server-rendered Health Card UI (.cshtml + Bootstrap, minimal JS)
- Chart.js: radar/spider chart (6-dimension score pillars), line chart (12-month score trend), doughnut chart (score band indicator)

DPI Integrations

- Account Aggregator (AA) via Finvu / FinFactor | Unified Lending Interface (ULI) DSP API
- OCEN 2.0 | GST System API | Udyam Registry | EPFO REST API (planned)

Estimated implementation cost (optional)

22 L

One-time setup (build + security)

31 L / yr

Recurring (cloud + data + support)

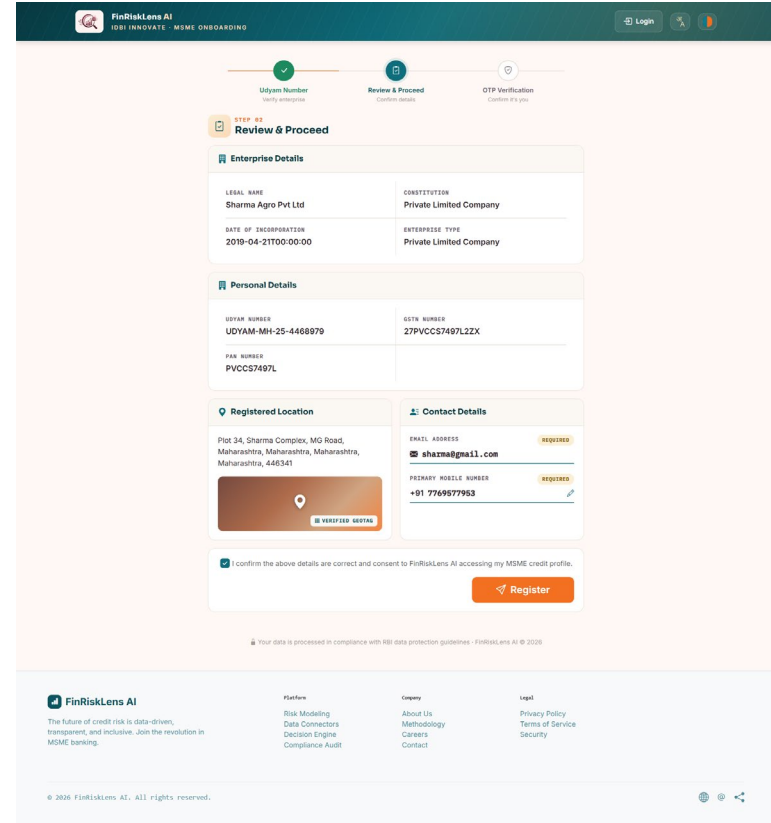
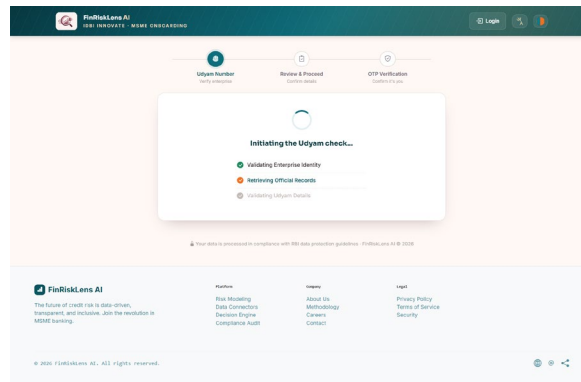
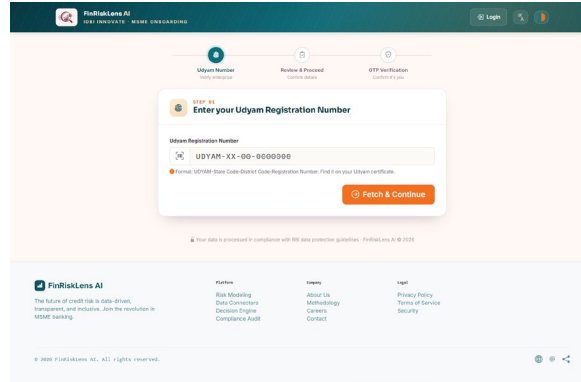
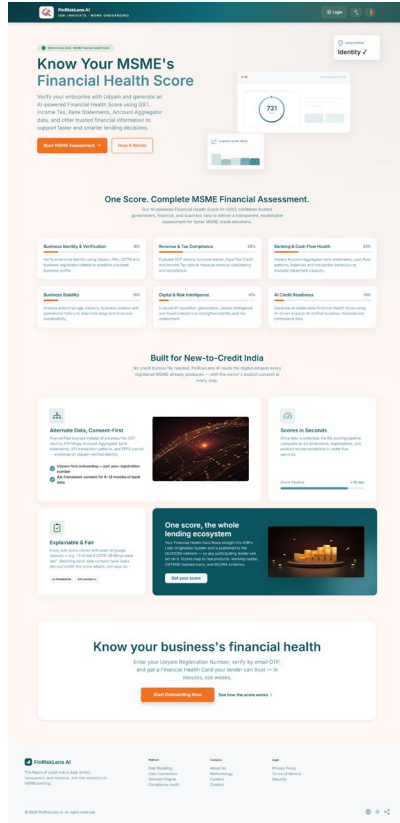
~ 1.15 Cr

3-year total cost of ownership

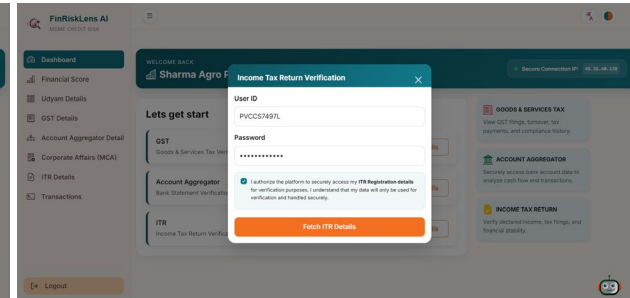
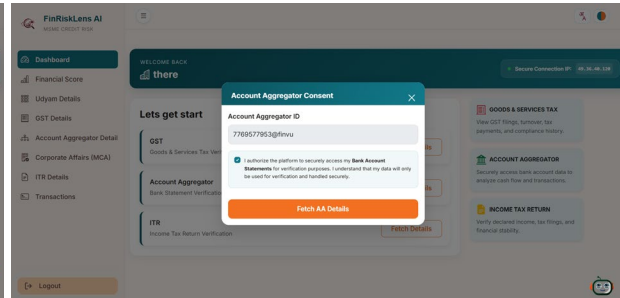
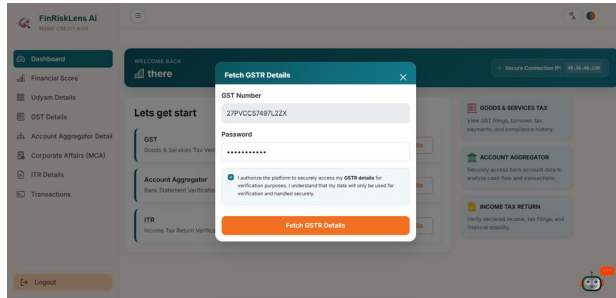
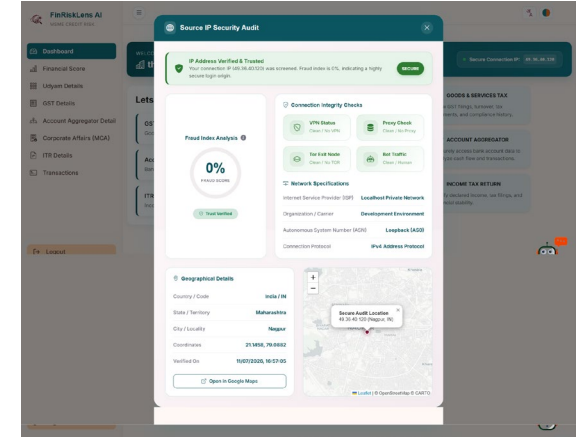
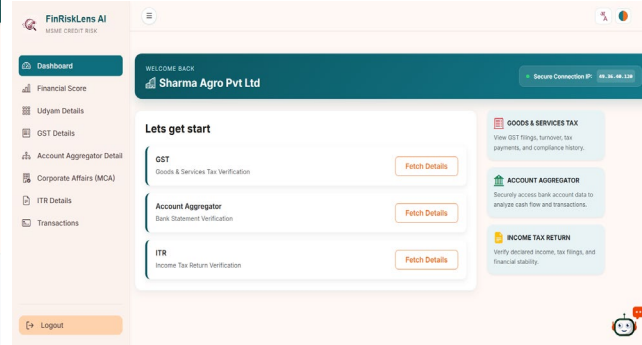
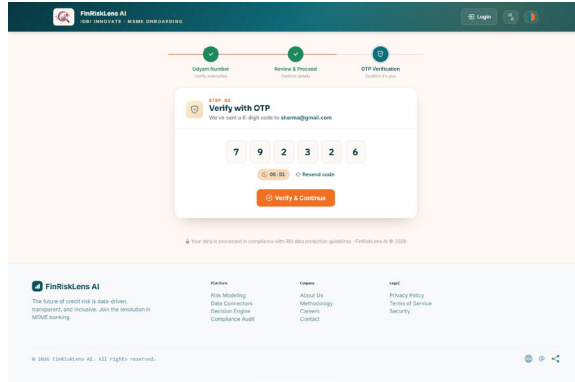
Cost breakdown - centralized cloud deployment serving all IDBI branches pan-India

Cost head	Type	Estimate (INR)
Development - 3-member team, ~6-month build	One-time	18 L
Security testing & RBI / DPDP compliance audit	One-time	4 L
Cloud & servers - Azure App Service, SQL, Blob, APIM	Annual	12 L
Data & API usage - AA (Finvu), GST, ITR	Annual	10 L
Maintenance, support & ML model retraining	Annual	9 L

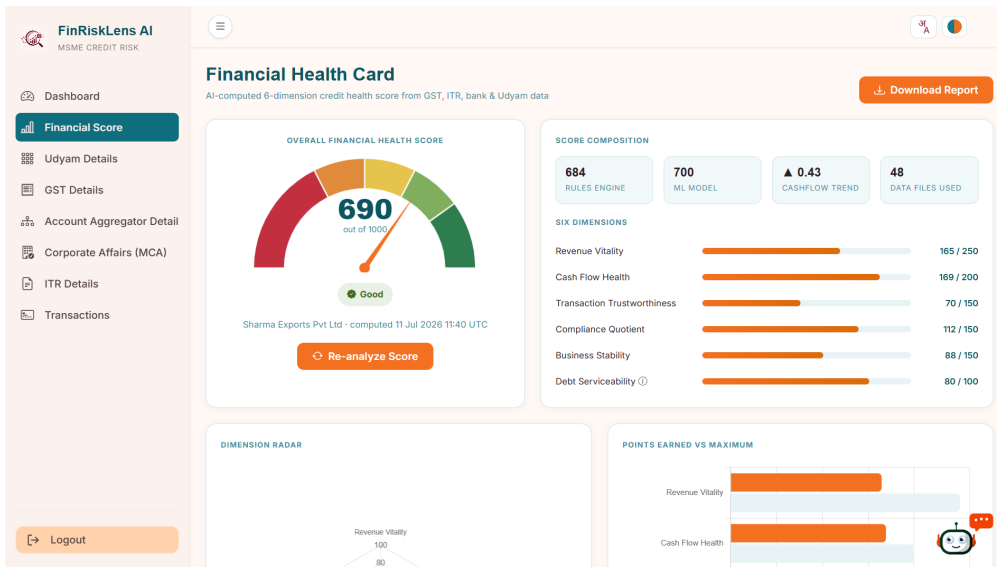
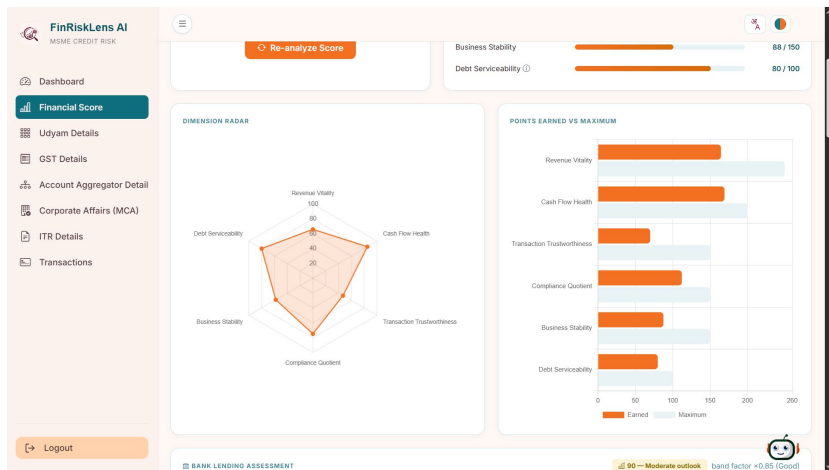
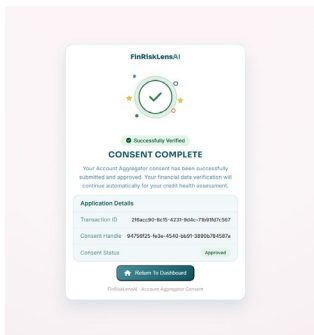
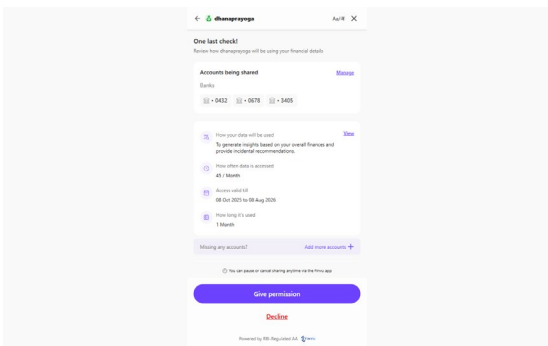
Snapshots of the prototype



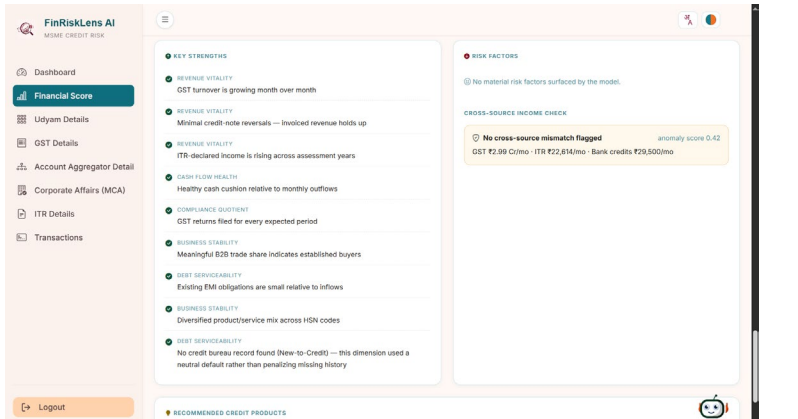
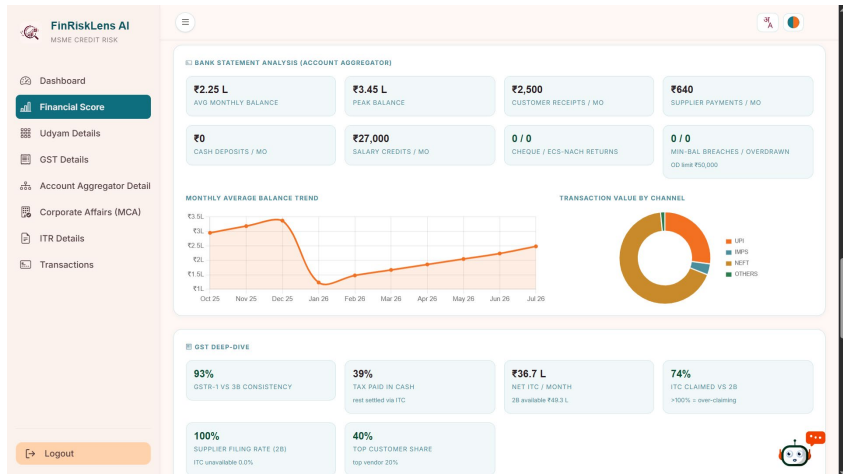
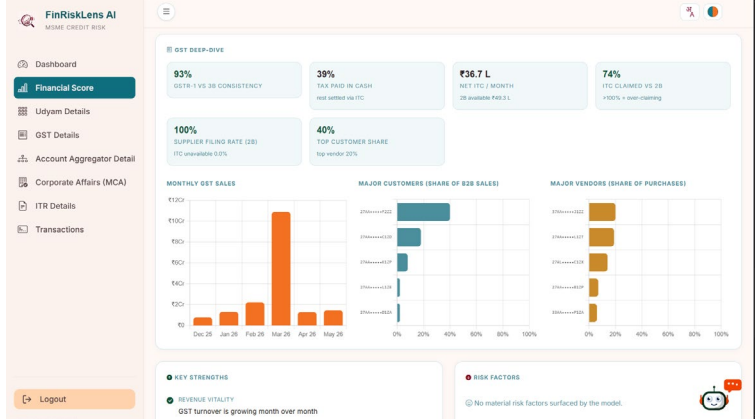
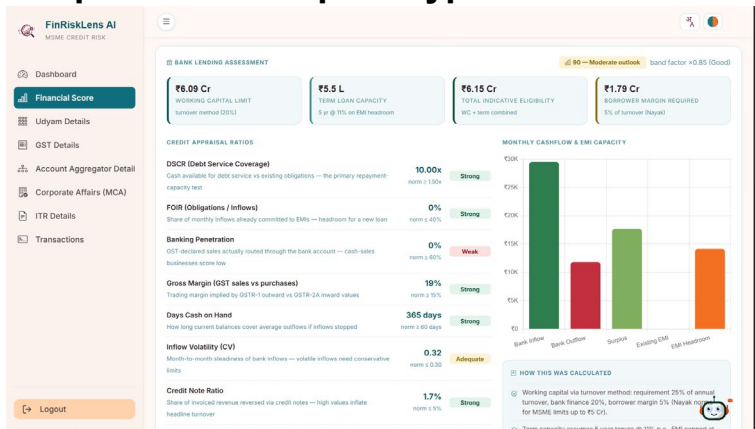
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype

FinRiskLens AI

MISME CREDIT RISK

Dashboard

Financial Score

Udyam Details

GST Details

Account Aggregator Detail

Corporate Affairs (MCA)

ITR Details

Transactions

REVENUE VITALITY

ITR-declared income is rising across assessment years

GST ₹2.99 Cr/mo - ITR ₹22,814/mo - Bank credits ₹29,500/mo

CASH FLOW HEALTH

Healthy cash cushion relative to monthly outflows

COMPLIANCE QUOTIENT

GST returns filed for every expected period

BUSINESS STABILITY

Meaningful B2B trade share indicates established buyers

DEBT SERVICABILITY

Existing EMI obligations are small relative to inflows

BUSINESS STABILITY

Diversified product/service mix across HSN codes

DEBT SERVICABILITY

No credit bureau record found (New-to-Credit) — this dimension used a neutral default rather than penalizing missing history

RECOMMENDED CREDIT PRODUCTS

Standard Working Capital Loan

₹5.38 Cr — ₹10.75 Cr

Good financial health score supports a standard working capital facility

Business Term Loan

₹7.17 Cr — ₹14.34 Cr

Consistent revenue and compliance history support term lending

Logout

FinRiskLens AI

MISME CREDIT RISK

Dashboard

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Transactions

GST RETURNS VERIFIED

AI-Powered GST Intelligence Analysis

Advanced GSTR-2B & GSTR-3B Business Intelligence Dashboard - Sharma Agro Pvt Ltd

GSTRN

PAN

UDYAM REGISTRATION

BUSINESS AGE

INDUSTRY

ENTERPRISE TYPE

GST HEALTH SCORE

94

Excellent GST Compliance

GST HEALTH SCORE

94

GST COMPLIANCE

98%

BUSINESS GROWTH

87%

RISK SCORE

Low

COMPLIANCE RATING

A+

GST HEALTH

Healthy

BUSINESS STABILITY

94%

CASH FLOW

Healthy

AI Business Assessment Engine

AI ASSESSMENT RATIONALE

Business has maintained excellent GST compliance 98%+ for the last 24 months.

Revenue growth is highly consistent (+18% outward supply YoY).

Vendor dependency is low (highest customer)

STABLE PERFORMANCE

GST Compliance & Filing History

REGISTRATION ACTIVE

9-MONTH FILING TIMELINE (GSTR-1 & 3B)

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sept

Oct

Nov

Dec

RETURNS

LATE

COMPLIANCE

REGISTRY

Logout

FinRiskLens AI

MISME CREDIT RISK

Dashboard

Financial Score

Udyam Details

GST Details

Account Aggregator Detail

Corporate Affairs (MCA)

ITR Details

Transactions

Overview

Directors

Charges

Risk Analysis

Corporate Profile

Company Name

Sharma Exports Pvt Ltd

CIN

U85074MH2021PTC524370

Type

Private Company

Category

Company limited by Shares

Status

Active

Listed

No

Incorporated

16 Mar 2021

ROC

RoC-Maharashtra

AI Observations

Company is Active

Private Limited Company

Registered under RoC-Maharashtra

No CIRP proceedings found

Company has active secured borrowings (8 open charges)

Proceed with GST & Banking assessment

Incorporation Journey

Company Incorporated

16 Mar 2021

RoC-Maharashtra

Directors Appointed

2 directors added

First Charge Created

2022

Secured borrowing initiated

Charges Modified

2021 - 2026

Multiple modifications

Current Status: Active

Today

Company operational

Quick Risk Indicators

COMPANY

Active

LISTED

No

CIRP

No

SUSPENDED

No

OPEN CHARGES

8

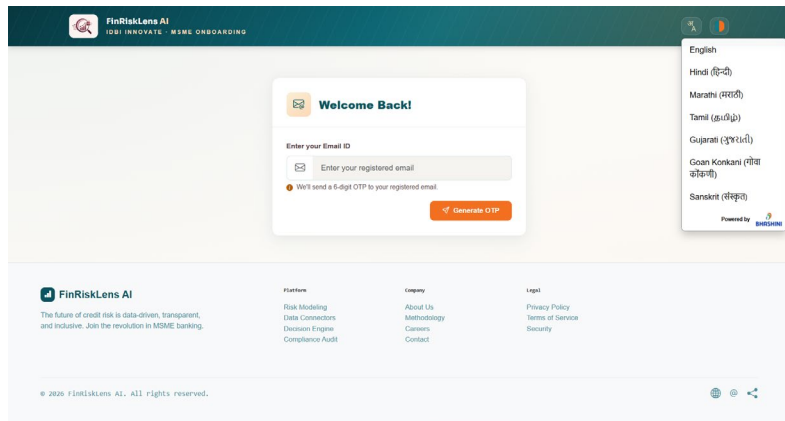
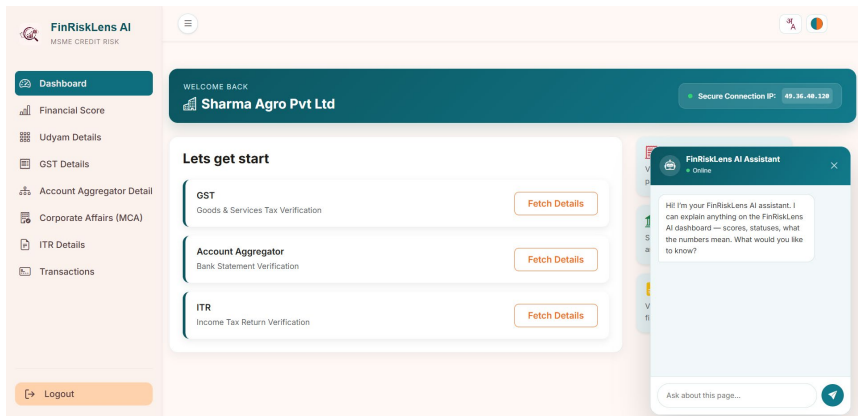
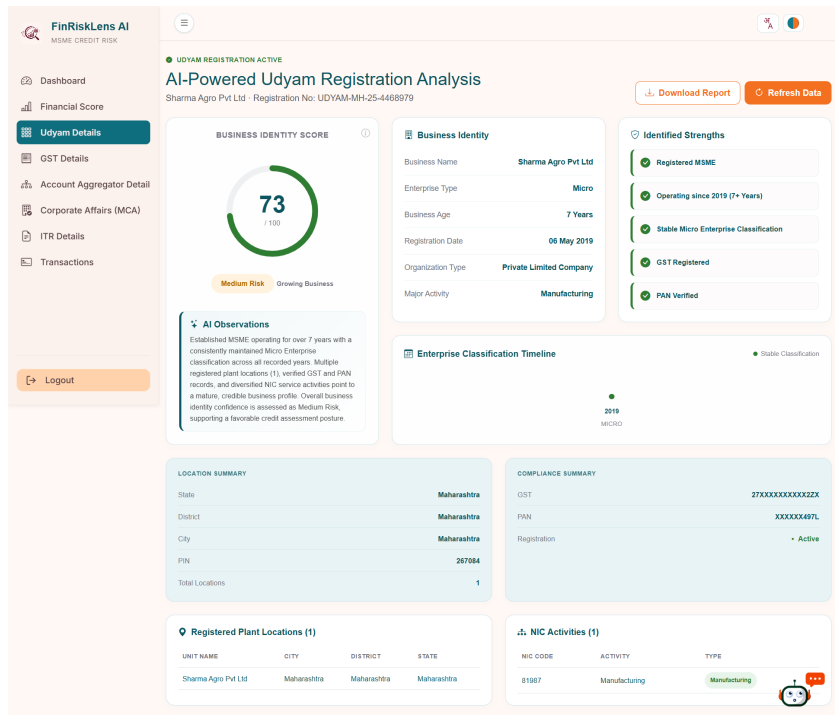
DIRECTORS

Stable

Corporate creditability is Low (85/100). Active company with 2 verified directors and secured financial charges (₹16.85 Cr open). Suitable for further GST and Banking assessment.

Logout

Snapshots of the prototype



Prototype Performance report/Benchmarking

719/1000

Live score on real 6 to 12mo GST data (band: Good)

< 10 s

End-to-end scoring (sub-second when warm)

4

DPI data sources fused: Udyam - GST - ITR – AA (+1 - EPFO)

3000

MSME profiles the model trains on

Benchmark vs. traditional credit assessment

Metric	Traditional / bureau	FinRiskLensAI
Turnaround time	2-3 weeks (manual)	Seconds (automated)
NTC / NTB borrowers	Rejected - no history	Scored on alternate data
Data sources	1 - bureau file	4 - Udyam, GST, ITR, AA (+1 - EPFO)
Explainability	Black-box score	PFI reason per dimension
Fraud / integrity check	None	3-way GST-ITR-bank anomaly

Additional Details/Future Development (if any)

Deeper Data Integrations

- MCA API - company / director & charge-registry checks
- EPFO payroll + GST e-invoice / e-way bill signals
- CERSAI - existing collateral / security-interest lookup

AML, Fraud & Compliance

- Sanctions / PEP / adverse-media name screening
- RBI Central Fraud Registry & willful-defaulter match
- FIU-IND suspicious-transaction flags on AA data
- AI-Powered Financial Insights Assistant with more features

Lending Ecosystem

- Full ULI DSP + OCEN 2.0 certification & go-live
- CIBIL / CRIF commercial-bureau enrichment (NTB)
- GST-invoice financing & co-lending workflows

Product & Intelligence

- Score trend history + PDF Health Card export
- Portfolio early-warning analytics for IDBI teams
- Vernacular AI assistant + continuous MLOps retraining

Provide links to your:

- GitHub Public Repository : <https://github.com/abhinavmukwane/FinRiskLensAI>
- Demo Video Link (3 Minutes): <https://video.finrisklensai.com> OR [Google Drive Link](#)
- Final Product Link: <https://finrisklensai.com>

Udyam Number for Demo Testing:

UDYAM-MH-20-0067888

UDYAM-MH-20-0067111

UDYAM-MH-20-0067394

UDYAM-GJ-01-0275433



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Thank you!